Blending self-defecting modal agents with FairBot: exact laws

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1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL: p means "the opponent cooperates with me" and q "the opponent cooperates with DefectBot". FairBot is $\Box p$, graded FairBots are $F_k = \Box^k p$. An agent is *unexploitable* if no modal opponent ever ends up defecting while the agent cooperates. Outcomes are computed in the height semantics: a game is a sequence of moves at heights $0, 1, 2, \ldots$, and its result is the eventual move.

Earlier findings on this site describe infinite families of unexploitable agents that cooperate with FairBot and CooperateBot but defect against their own copy: the gate families $V_{n,r} = \Box(\Box^n \bot \vee (p \wedge (q \vee \neg\Box^r q)))$, with $W_n = V_{n,1}$, and $J_{a,b,j} = \Box^{a+1}(\Box^b p \wedge (q \vee \Box(\Box^j \bot \vee \neg q)))$. Here we ask what happens when such an agent is *blended* with another agent X: a formula σ_S without atoms picks out a set S of heights (finite, or all but finitely many), and the blend plays like X at the heights in S and like the family member elsewhere. X is FairBot, a graded FairBot, DefectBot or CooperateBot.

Result

- **FairBot-type blends.** For a graded-FairBot blend with finitely many selected heights: - a W_n blend cooperates with itself exactly when S meets $\{k, \ldots, n-1\}$; - a $J_{a,b,j}$ blend exactly when $k \le j$ and S contains all of $\{j, \ldots, a+b\}$; - for $V_{n,r}$ the condition is a window determined by the blend's DefectBot trace, and it is not monotone: for $V_{3,2}$, selecting height 2 makes the blend cooperate with itself, while selecting heights 2 and 3 does not.

All these blends stay unexploitable and keep cooperating with FairBot and CooperateBot.

- **DefectBot blends.** Playing DefectBot at even one height rules out mutual cooperation with FairBot and with every F_k, because an agent ends

in mutual cooperation with F_k exactly when it cooperates at every height of that game. A DefectBot blend of W_n still cooperates with itself exactly when S is a non-empty subset of $\{0, \ldots, n-1\}$; for $J_{a,b,j}$ never.

- **CooperateBot blends** cooperate with themselves only when CooperateBot is played at all but finitely many heights, and then DefectBot exploits them.
- **Not FairBot in disguise.** For finitely many selected heights, a single graded FairBot (F_{j+1} for J, $F_{\max(1,n-r+1)}$ for V) behaves differently against the blend than against FairBot or any $\Box^m p$. A blend selecting all but finitely many heights behaves exactly like FairBot only in the trivial case (FairBot at every height from 1 on); otherwise an explicit opponent tells them apart.

Why it matters

Unexploitable agents that defect against themselves seem to sit far from FairBot, which cooperates with itself. These laws say exactly how much FairBot behaviour, and at which heights, is needed to restore self-cooperation without losing safety, and they give an infinite supply of unexploitable self-cooperating agents that are provably not FairBot.

Proved vs checked

All laws are proved for all indices in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [29], cited, not formalised (see also [12]). Finite grids of cases were also checked independently by GL fixed-point solving with certificates.

Novelty statement

We believe this result is new. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire, Christiano, Fallenstein, Herreshoff, Yudkowsky, "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic" (arXiv 1401.5577): unexploitability (Section 2), rank and Theorem 4.10 (Section 4), WaitFairBot_K and the obstacle theorem (Section 5).
- LessWrong, "Is there only one FairBot?" (transhumanist_atom_understander, 29 Aug 2026) and comments: Zac Hatfield-Dodds shows the uniqueness conjecture holds up to modal depth 1 for rank-0 agents and fails from depth 2 (Box Box p; FairBot over PA+1 = Box(p or Box False); Box^k p for k >= 3); PrudentBot separates Box Box p from FairBot; the author notes Box Box p is not provably unexploitable in PA. No enumeration or counts.

- LessWrong / Alignment Forum wiki "Modal combat": DefectBot, CooperateBot, FairBot, PrudentBot, CliqueBot, TrollBot; no uniqueness result, no enumeration.
- LaVictoire et al. 2014 (MIPC); comment thread on "Robust Cooperation in the Prisoner's Dilemma" (LessWrong, on instability of FairBot equilibria); Hobson, "Yet More Modal Combat" (2021); Critch, parametric bounded Löb (arXiv 1602.04184); Critch, Dennis & Russell (arXiv 2208.07006); MIRI provability repository (github.com/machine-intelligence/provability, a modal-agent evaluator, no published census).
- Out of scope by direction: probabilistic blends (Oesterheld arXiv 2211.05057; Cooper et al. arXiv 2412.14570).
- Web searches for deterministic blends, behavioural distance and classifications of modal agents (2026-10-20/21); none means not found in this search.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result extends Barasz et al. (arXiv 1401.5577): FairBot, the height guards of WaitFairBot_K, and the evaluation of modal agents on linear GL frames; Boolos, The Logic of Provability (1993): semantics of GL; Our earlier published findings on W_n and the gate families. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire, Christiano, Fallenstein, Herreshoff, Yudkowsky, "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic" (arXiv 1401.5577): unexploitability (Section 2), rank and Theorem 4.10 (Section 4), WaitFairBot_K and the obstacle theorem (Section 5).
- LessWrong, "Is there only one FairBot?" (transhumanist_atom_understander, 29 Aug 2026) and comments: Zac Hatfield-Dodds shows the uniqueness conjecture holds up to modal depth 1 for rank-0 agents and fails from depth 2 (Box Box p; FairBot over PA+1 = Box(p or Box False); Box^k p for k >= 3); PrudentBot separates Box Box p from FairBot; the author notes Box Box p is not provably unexploitable in PA. No enumeration or counts.
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- Web searches for deterministic blends, behavioural distance and classifications of modal agents (2026-10-20/21); none means not found in this search.
- Boolos, The Logic of Provability (1993): GL semantics.

If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was checked by scripts that run Lean proofs; the proof and check files are included so anyone can re-run them.

2 Claims

Claim K14

For $n \geq 2, k \geq 1$, an F_k height blend of W_n with finite S self-cooperates iff $S \cap \{k, \ldots, n-1\} \neq \varnothing$; all such blends remain unexploitable and cooperate with FairBot and CooperateBot.

Status: derived ů author check only.

Verified.

- `checks/K14-graded_self.py`: Six general Lean theorems: first-hole self-play law, DefectBot trace, first failure existence, early-hole criterion, FB and CB partners; SHA-pinned source, no sorryAx, three mutations rejected. — pass (Lean-proved, no sorryAx)
- `checks/K14-graded_grid.py`: 144 GL-certified cells for n2,3; k1,2,3; eight height selectors; self, FairBot, CooperateBot; wrong-candidate controls. — pass
- `checks/K14-blend_gate.py`: General graded-FairBot blend unexploitability and iterated-box semantics (K7). — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K6

For $n \geq 2$, a FairBot height blend of W_n self-cooperates iff the first false height t of its DefectBot trace satisfies $t < n$ or all $h > t$ belong to S; all such blends remain unexploitable.

Status: derived ůa author check only.

Verified.

- `checks/K6-fb_self.py`: General Lean self-play iff law; first-hole and DefectBot trace semantics; first failure exists; early-hole criterion; all-height FB/CB cooperation; no sorryAx; three mutations rejected. — pass (Lean-proved, no sorryAx)
- `checks/K6-fb_grid.py`: All 48 GL-certified self/FB/CB cells for n=2,3 and eight selectors arbitrary on heights 0,1 and constant from 2; wrong-candidate controls. — pass
- `checks/K14-blend_gate.py`: General Lean unexploitability of all graded-FairBot blends, including FairBot. — pass

Evidence.

- reasoning: The q gate has exactly one hole, at the first false height of the DefectBot trace. Either branch forces p(g) for n\<=g\<h, preserving eventual unexploitability. General Lean proof pending.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K16

For an F_k height blend of $V_{n,r}$, let d be its DefectBot trace, t its first false height and $E = \{g \geq n : \neg d(g),\ g < t + r\}$. Self-play cooperates iff every height above every $g \in E$ is selected; for finite S, iff $E = \varnothing$.

Status: derived ůa author check only.

Verified.

- `checks/K16-v_self.py`: 11 general Lean core theorems: obstruction law, first-hole window, finite-set law, singleton iff, DB trace, FB/CB cooperation, universal unexploitability; SHA-pinned, no sorryAx; 3 mutations rejected. — pass (Lean-proved, no sorryAx)
- `checks/K16-v_grid.py`: 486 GL-certified cells including all 480 grid cells for five (n,r) pairs, k1,2, 16 selectors vs self/FB/CB and six nonmonotonicity headline cells; wrong candidate controls. — pass
- `checks/K16-v_reestablished.py`: SHA-pinned full prior V-family script re-executed: nine Lean theorem checks, five rejected mutations, literal tables and 35 separated family members. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K19

For $1 \leq j \leq a + b$ and $k \geq 1$, an F_k height blend of $J_{a,b,j}$ with finite S self-cooperates iff $k \leq j$ and $\{j, \ldots, a + b\} \subseteq S$; all such blends are unexploitable and cooperate with FairBot and CooperateBot.

Status: derived ů author check only.

Verified.

- `checks/K19-j_self.py`: 9 SHA-pinned Lean core theorems: literal J normalization, general obstruction law, finite interval law, DB trace, universal unexploitability, FB/CB cooperation; no sorryAx; 3 rejected mutations. — pass (Lean-proved, no sorryAx)
- `checks/K19-j_grid.py`: 581 GL-certified cells: exhaustive six-triple/two-k/16-selector self-FB-CB grid and 5 headline cells including J111 selection1 versus2 versus1,2; F2 separator; wrong-candidate controls. — pass
- `checks/K19-j_reestablished.py`: Full SHA-pinned prior J-family script, including G and H special cases: literal tables, 70 distinct agents, formula-string cells, re-executed. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K9

A DefectBot height blend of W_n ($n \geq 2$) self-cooperates iff $\varnothing \neq S \subseteq \{0, \ldots, n - 1\}$; a CooperateBot blend self-cooperates iff S is cofinite.

Status: derived ů author check only.

Verified.

- `checks/K9-other_self.py`: Lean core proofs of the general DB and CB self-play iff laws and both DefectBot-trace formulas; SHA-pinned source; no sorryAx; three mutations rejected. — pass (Lean-proved, no sorryAx)
- `checks/K9-other_grid.py`: All 32 GL-certified self-play cells for DB/CB blends, n=2,3, eight selectors arbitrary at 0,1 and constant from 2; wrong-candidate controls. — pass

Evidence.

- reasoning: DB selection below n repairs the q gate without retained self-defects; selection at or above n propagates failure. Finite CB selection leaves a first DB-trace hole at or above n+1 and cannot shield it forever. General Lean proof pending.

Prior art. Partial prior art.

Claim K25

For $r \geq 1$, a DefectBot height blend of $V_{n,r}$ self-cooperates iff $S \subseteq \{0, \ldots, n-1\}$ and some $s \in S$ satisfies $s + r \leq n + 1$; all remain unexploitable. CooperateBot blends self-cooperate iff S is cofinite, when DefectBot exploits them.

Status: derived ů author check only.

Verified.

- `checks/K25-v_other.py`: Six SHA-pinned Lean core theorem groups: general DB iff, CB iff eventual selection, DB traces, DB safety, finite CB safety, cofinite CB exploitation; no sorryAx; three rejected mutations. — pass (Lean-proved, no sorryAx)
- `checks/K25-v_other_grid.py`: 896 GL-certified self/DB cells for 14 n,r pairs, 16 selectors, DB/CB branches; wrong-candidate controls. — pass
- `checks/K16-v_reestablished.py`: SHA-pinned prior V-family literal semantics and Lean results re-established as in K16. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K23

For $1 \leq j \leq a + b$, no DefectBot height blend of $J_{a,b,j}$ self-cooperates; a CooperateBot blend self-cooperates iff its selector is cofinite, and then DefectBot exploits it. Every DefectBot blend and every finite-selector CooperateBot blend is unexploitable.

Status: derived ů author check only.

Verified.

- `checks/K23-j_other.py`: Six SHA-pinned Lean core theorems for arbitrary indices/selectors: traces, no DB self-cooperation, CB self iff eventual selection, DB safety, finite CB safety, cofinite CB exploitation; no sorryAx; three rejected mutations. — pass (Lean-proved, no sorryAx)
- `checks/K23-j_other_grid.py`: All 384 self/DB cells GL-certified for six J triples, 16 selectors and DB/CB branches; wrong-candidate controls. — pass
- `checks/K19-j_self.py`: K19 literal J normalization into the height gate, SHA-pinned Lean proof. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K28

For every agent and $k \geq 1$: it ends CC with $F_k = \Box^k p$ iff it cooperates at every height of that game. Hence any height blend that plays DefectBot at some height, whatever its other branches, never ends CC with FairBot or any F_k; if unexploitable, it ends DD.

Status: derived ů author check only.

Verified.

- `checks/K28-c046_mixed.py`: Lean 4 core fk_cc_iff, fk_defects, db_height_blocks (SHA-pinned, no sorryAx, 3 mutations rejected); literal evaluator 6804 mixed F_k/DB/V_{n,r} blends (n1..4, r=1..n+1, k1,2, 243 labelings) vs self, F_1..F_3, CB: CC with F_m iff all-height cooperation; any DB label gives DD and F_m defects from s+m; DB-free blends match K22; 702 DB-labelled self-CC blends, none FB-CC — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: F_k at h reads the agent's moves at all g\<=h-k, so a single defection at s makes F_k defect from s+k on; a DefectBot-selected height is such a defection regardless of the other branches.

Prior art. Partial prior art.

Claim K20

For finite S, $k, m \geq 1$: the F_k height blend of $J_{a,b,j}$ and $F_m = \Box^m p$ cooperate at every height iff $m \leq j$, else both eventually defect; so F_{j+1} separates every such blend, including every self-cooperating one, from FairBot and from each $\Box^{m'} p$.

Status: derived ů author check only.

Verified.

- `checks/K20-c046_j_xcheck.py`: Lean 4 core fm_db, fm_coop (m<=j: all-height CC, any S), fm_defect (j<m, finite S: eventual DD), fm_fm, separator; SHA-pinned, no sorryAx, 3 mutations rejected; exact engine: 53760 finite-S cells vs F_m (m<=7) CC iff m<=j; 1668 finite self-CC blends none FB-equivalent; K19 laws on 15360 blends — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: F_m's DefectBot trace is true exactly below m, so J's q-gate is open everywhere iff m\<=j; otherwise it closes at g=m, and with finite S an unselected height past it defects, after which F_m and then both blend branches defect forever.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K22

For $r, k, m \geq 1$ and any S, the F_k height blend of $V_{n,r}$ ($W_n = V_{n,1}$) and $F_m = \Box^m p$ cooperate at every height iff $m + r \leq n$ or all $h > \max(n, m)$ are selected, else both eventually defect; for finite S, $F_{\max(1,n-r+1)}$ separates the blend from every $\Box^{m'} p$.

Status: derived ů author check only.

Verified.

- `checks/K22-c046_v_fm.py`: Lean 4 core vfm_coop, vfm_defect, vfm_law (iff for arbitrary S), fm_fm, v_separator; SHA-pinned, no sorryAx, 3 mutations rejected; exact engine: 55296 cells (6912 V blends n2..5, r=1..n+1, k1..3, 128 selectors vs F_1..F_8) match the law; all blends unexploitable; 1010 finite self-CC blends DD with F_max(1,n-r+1), none FB-equivalent; hand examples W2{1}, V32{2}, W4{2} — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Against F_m the q input is (g\<m), so the V gate fails at g\>=n exactly for m\<=g\<m+r; an unselected height above the lowest hole max(n,m) defects, after which F_m and both blend branches defect forever; otherwise strong induction gives all-height CC.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K24

An F_k height blend of $J_{a,b,j}$ ($k \geq 1$, cofinite S) is FairBot-equivalent iff $k = 1$ and all $h \geq 1$ are selected; else $R_h(q)$ (its DefectBot trace true at $h \geq 1$) or Y_{h_0} (unselected $h_0 \geq a + b + 1$) gets CC or DC from it and DD from FairBot.

Status: derived ů author check only.

Verified.

- `checks/K24-c046_jcof.py`: Lean 4 core db_trace, reader, cases3, fb_case, rq_blend, rq_fb, y_blend, y_fb (arbitrary a,b,j, k>=1, cofinite S); SHA-pinned, no sorryAx, 3 mutations rejected; exact engine on 7680 cofinite J blends: FB-equivalent iff k=1 and {1..5} in S (80), 7344 R_h(q)-separated

(CC vs DD), 256 Y_h0-separated (DC vs DD); 6288 self-CC, 6208 separated incl. all 108 bank-unseparated; G11 hand example — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Cofinite blends differ from FairBot either in the DefectBot trace at some h\>=1 (read by R_h(q)) or at an unselected height h0\>=a+b+1 where J ignores p at h0-1 (probed by Y_h0, which defects there only against partners that already cooperated); otherwise the blend is the box-p operator.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K26

For $n, r, k \geq 1$ and any finite or cofinite S, the F_k height blend of $V_{n,r}$ ($W_n = V_{n,1}$) is FairBot-equivalent iff $k = 1$ and all $h \geq 1$ are selected; else, for cofinite S, $R_h(q)$ gets CC from it and DD from FairBot, or Z_{n,h_0} gets DD from it and CC from FairBot.

Status: derived ů author check only.

Verified.

- `checks/K26-c046_vcof.py`: Lean 4 core db_trace, reader, cases3, fb_case, rq_blend (any unselected branch), rq_fb, z_blend, z_fb; SHA-pinned, no sorryAx, 3 mutations rejected; exact engine on 7680 V blends (n1..5, r=1..n+1, k1..3, 64 masks x finite/cofinite tail): FB-equivalent iff k=1 and S contains 1..5 cofinitely (40); 3664 R_h(q) CC/DD, 136 Z_{n,h0} DD/CC; 2522 cofinite self-CC, 2482 separated; W2 hand example unexploitable, self/FB/CB CC — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: A cofinite blend differs from FairBot either in its DefectBot trace at some h\>=1 (read by R_h(q)) or, with FairBot's DefectBot trace, at an unselected h0\>n where the V q-gate can be closed by an opponent that defects only against DefectBot at height n; otherwise the blend is the box-p operator. Finite S is K22.

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261020-1600`, claims K14, K6, K16, K19, K9, K25, K23, K28, K20, K22, K24, K26, run-folder commit `e3d59e30bce37b58223f480d4f0d5a79bbff59ce`, published 2026-10-21.

3 Discussion

The blends are a deliberately simple, deterministic way to mix agents: a formula without atoms decides, height by height, which agent to imitate. Probabilistic mixtures of FairBot-like agents have been studied elsewhere [25] and are not covered here. The results are stated for the W, V and J families; whether similar window laws hold for every self-defecting unexploitable agent is open. The use of height guards to switch behaviour goes back to WaitFairBot in Barasz et al.

4 How to verify

Each finding has its check scripts in `claims/blend-window-laws/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

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